Project: Edge Detection in Noisy Images Using 2D Filtering

ECE 0402: Signals, Systems & Probability

Heyang Zhang (Stefan) Zhang

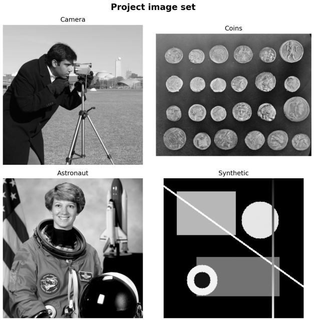
Sichuan University - Pittsburgh Institute

Problem and Motivation

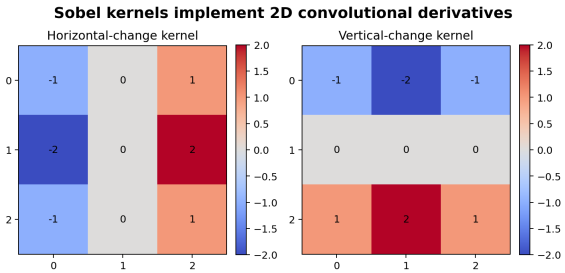
- Images are 2D signals, so edge detection is a filtering problem.
- Noise and true edges both trigger large derivative responses.
- The main question: how does filtering improve edge detection under noise?

Why Edges Are High-Frequency Structure

- An edge is a rapid intensity change.
- A derivative-based operator amplifies rapid changes.
- Noise also changes rapidly, so naive edge detection confuses noise with structure.

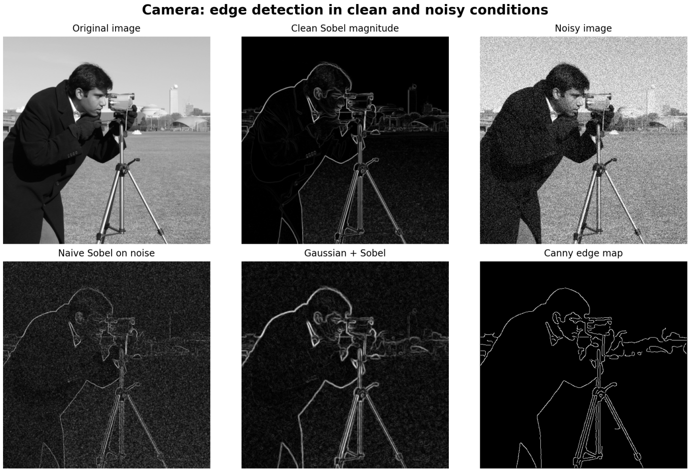


2D Convolution and Sobel Intuition



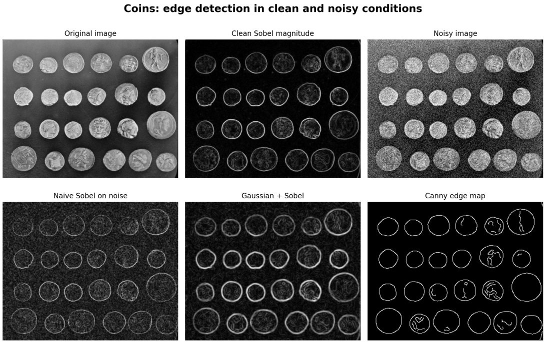
- Sobel uses two convolution kernels.
- One estimates horizontal change.
- One estimates vertical change.
- The gradient magnitude highlights boundaries.

Failure Case: Noise Corrupts Edges

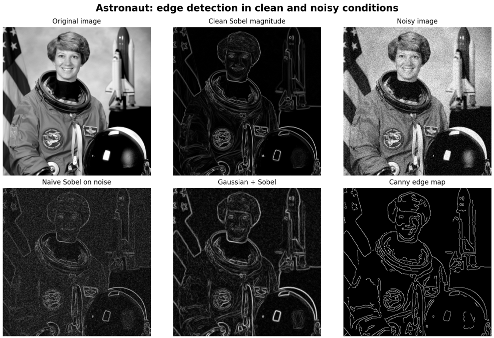


Fix: Smooth Before Differentiating

- Gaussian smoothing suppresses unstable high-frequency noise.
- Sobel after smoothing recovers stronger large-scale boundaries.
- Too much smoothing also erases weak detail.

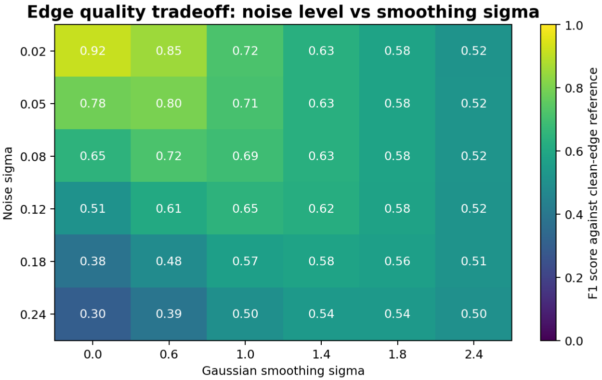


Canny as a Refined Pipeline



- Canny combines smoothing, gradients, suppression, and thresholding.
- It is stronger than raw Sobel.
- But it still depends on the same filtering logic.

Tradeoff Heatmap



Higher noise usually needs more smoothing, but oversmoothing removes useful edges.

CNN and Diffusion Connection

- Early CNN filters often learn edge-like patterns.
- Diffusion models repeatedly remove noise from an image estimate.
- Neither idea replaces classical filtering intuition.
- They extend it.

Conclusion

- Edge detection is a filtering problem, not a black box.
- Noise breaks naive derivatives because both noise and edges are high-frequency.
- Pre-smoothing improves robustness, but detail is the cost.
- The same logic carries forward into modern vision systems.

References

1. ECE 0402 course project guideline.
2. scikit-image: Canny edge detector.
3. scikit-image API: filters.
4. HIPR2: Sobel Edge Detector.
5. HIPR2: Canny Edge Detector.
6. B. P. Lathi, *Linear Systems and Signals*.
7. Luis F. Chaparro, *Signals and Systems Using MATLAB*.

Team Responsibilities

Member	Responsibility	Concrete Output
Member A	Framing and references	Topic definition, slide story
Member B	Implementation	Python modules, figures
Member C	Validation	Heatmap analysis, checks
Member D	Delivery	Slide edits, narration, rehearsal